RESEARCH ARTICLE

Evidence before prediction: a configuration-driven framework for testing how molecular descriptors shape property models

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Abstract

Background. When a chemist builds a machine-learning model for a molecular property, most of the attention goes to the learner: a graph neural network, a descriptor-based model, or a hybrid. In practice much of the outcome is decided earlier, by the quality of the data, by which of several hundred descriptors enter the model, and by the role those descriptors are given. These choices are usually buried in task-specific scripts, so a reader cannot tell which were tested, which were assumed, and which produced the reported score.

Results. We present `chemistry-framework`, an open-source workflow in which an experiment is declared in one validated configuration file and every run writes a record that can be replayed. We tested three hypotheses. *H1, that training-partition evidence can guide descriptor choice*, is partly supported: the chosen subset had the better mean in all 20 dataset–budget cells and beat a 50-draw random control significantly in 16, but the rule ranked fourth of six selectors (mean rank 3.65). *H2, that the effect of adding descriptors transfers between implementations*, holds on four of five endpoints: the framework and Chemprop v2.3.1 agreed in direction and approximate size (+12.1% versus +19.7% on FreeSolv; at most about 2% elsewhere), and on ESOL one divergent run (2.76 RMSE) explained the disagreement: six retrainings span 0.75–6.12, and bounding the descriptor inputs collapses them to 0.65–0.72, where both place the effect at zero. *H3, that routes and predictive roles matter but are endpoint-specific*, holds: the full 210-descriptor hybrid led on ESOL and FreeSolv, the recommender route on Lipophilicity and graph-only on BACE, while on BBBP every route lay within 0.007 AUROC. No compact route separated from its matched random control.

Conclusions. Evidence from the training partition is useful for narrowing choices, but it does not name a universal winner, and a mean that hides one divergent run is unfinished evidence. Declaring the audit, the descriptor evidence, the route and the role in one configuration makes these comparisons cheap to run and possible to audit. The framework does not prescribe a better predictor; it makes the alternatives testable on the user's own data.

Keywords: Cheminformatics; molecular property prediction; feature selection; graph neural networks; QSAR; reproducibility; configuration-driven software

Background

A chemist who wants a model for a molecular property usually starts with a table: one row per compound, a SMILES string, and a measured value. Turning that table into a trustworthy prediction involves three decisions that are made before any model is fitted. The first is whether the data can carry the weight placed on them: are the structures valid, are they duplicated, do the labels agree, and does the training set cover the chemistry that will be predicted? The second is which of several hundred computed descriptors should enter the model, if any. The third is the role those descriptors should play: passed to the model as additional input, used as a fixed baseline, or used as the starting point that a graph network corrects.

In practice these three decisions are encoded in task-specific scripts. The script mixes the chemical judgement with the engineering plumbing, so a reader cannot tell which options were tested, which were assumed, and which

Table 1. Emphasis of related open-source molecular machine-learning tools, summarised from the cited publications. Entries are not exhaustive feature lists.

Tool	Primary emphasis	Feature handling	Roles of feature groups
Chemprop v2 [49]	Efficient, modular D-MPNN training and inference	Optional extra atom-, bond-, and molecule-level features	Direct (concatenated with the learned representation)
DeepMol [53]	Automated search over preprocessing, representation, and model pipelines	Feature extraction and selection as pipeline steps	Direct model input
QSARtuna [54]	Automated QSAR model and hyperparameter optimisation	Choice among descriptor and fingerprint representations	Direct model input
QSPRpred [56]	Reproducible, serialisable QSPR modelling, including proteochemometrics	Training-only variance, correlation, and Boruta filters	Direct model input
This work	Auditable link between feature evidence and model configuration	Training-partition evidence scorecard, dataset audit, relevance and redundancy recommender	Direct, fixed prior, or learned residual, declared per feature block

produced the reported number. Six months later the comparison is hard to repeat exactly. The problem is not a lack of algorithms. Descriptor-based models, fingerprints, and graph neural networks are all well developed, and benchmark collections such as MoleculeNet and the Therapeutics Data Commons allow systematic comparisons across endpoints [8]. Prior work shows that descriptor-based models can match or outperform graph models on some tasks, and that adding descriptors to a learned graph representation can help on others. The recurring lesson is that the usefulness of a representation depends on the endpoint, the curation of the data, and the evaluation protocol, not on the model class alone.

Decisions about features are equally consequential. Quantitative structure–activity and structure–property relationship (QSAR/QSPR) guidance asks for a defined endpoint, a transparent algorithm, and a stated applicability domain. Feature selectors address part of this: filters rank descriptors by correlation or mutual information, while embedded methods use a fitted learner, such as Random Forest importance or sparse regression, to rank them. Every such ranking is evidence for a particular dataset, learner, and budget. It is not a universal statement about which molecular properties matter, and presenting it as one is a common source of over-claiming.

A related decision is how the selected information enters the predictor. Descriptors may be concatenated with a learned graph representation, used as a fixed linear prior learned on the training set, or used as the baseline that the graph model corrects. We call the assignment of a feature block to a task a *route*, and the way its output enters the predictor a *role*. These roles answer different modelling questions, yet they are usually implemented in separate scripts or frozen into a pipeline, which makes an otherwise simple comparison expensive.

Open-source software already supplies much of the infrastructure. Chemprop provides efficient directed message-passing networks [49], DeepMol and QSARtuna automate pipeline and model search [53, 54], QSPRpred emphasises reproducible and serialisable QSPR workflows [56], and ZairaChem targets automated screening cascades. General tools such as Hydra and MLflow record settings and runs. The gap addressed here is narrower and, we argue, more useful to a chemist: none of these tools makes training-partition feature evidence, descriptor recommendations, feature routes, and predictive roles explicit inside one auditable experiment record (Table 1).

We therefore built `chemistry-framework`, an open-source Python package in which an experiment is declared in one plain-text configuration file. The file names the dataset, the molecular representations, the split, the model, and the role of each descriptor block; it is checked against a typed schema before any computation starts. Fig. 1 shows the five stages a user goes through, from a table of SMILES and measured values to a saved record that another person can replay. The framework does not replace the chemist's judgement. It makes each judgement visible, testable, and cheap to change.

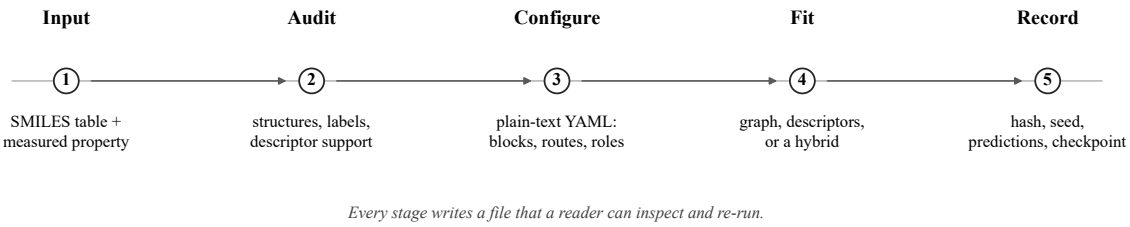


Fig. 1. The five stages of a `chemistry-framework` experiment. A table of SMILES and measured properties is read and audited, the experiment is declared in a validated configuration file, the selected representations are fitted, and every run writes a record (configuration hash, split seed, predictions, checkpoint) that a reader can inspect and re-run. The framework changes what is explicit, not what is chemically possible.

What this study tests

The paper is organised around three hypotheses, stated here so that the reader can judge the results against them rather than after the fact.

H1 — Evidence is usable. Training-partition diagnostics of descriptors, together with a transparent relevance-and-redundancy rule, carry enough information to guide descriptor choice. *Prediction:* (i) descriptor subsets chosen from this evidence should beat random subsets of the same size in most dataset–budget cells tested; (ii) the rule should rank in the upper half of the selectors compared against it; and (iii) the choice should be stable across splits. *Falsified if:* the evidence-based choice is indistinguishable from a random one, or if the rule ranks in the lower half of the selectors.

H2 — The descriptor effect transfers. The effect of adding descriptors to a graph model is a property of the endpoint and the data, not of one software implementation. *Prediction:* an independently written implementation (Chemprop v2.3.1, with a different encoder, featurisation and training loop) should change in the same direction and by a similar amount on most endpoints. Because the two systems differ in more than the descriptor block, agreement is evidence that the effect transfers, not a decomposition of it into architecture and features; the architecture-unconfounded estimate is the within-system contrast. *Falsified if:* the two implementations disagree in sign across endpoints.

H3 — Route and role matter, but not universally. Compact descriptor routes and alternative predictive roles change results, yet the best choice is endpoint-specific and small compared with split-to-split variation. *Prediction:* no single route or role wins everywhere, and most differences should be comparable to the between-split standard deviation. *Falsified if:* one configuration wins on every endpoint by a margin that exceeds the split variation.

Testing these hypotheses requires more than a new script. It requires the same splits, the same checkpoints, and the same evaluation rule across every model, and it requires that the comparison be reproducible by someone who did not run it. The remainder of the paper reports the design (Methods), the results for each hypothesis (Results), what they do and do not imply for a practising chemist (Discussion), and the conclusions (Conclusions). Table 2 maps each hypothesis to its experiment and to the section and table where the evidence is reported.

Results

Performance depends on the endpoint, not on the model class

The best model depended on the task (Tables 3 and 4). The descriptor MLP and the framework direct hybrid tied for the lowest mean RMSE on ESOL (0.623), the framework direct hybrid was best on Lipophilicity (0.606), and Chemprop's

Table 2. Reading map: each hypothesis, the experiment that tests it, and where the answer is reported.

	Hypothesis	Experiment	Reported in
H1	Training-partition evidence guides descriptor choice	Scorecard and dataset audit on five endpoints; six selectors at four budgets with a Random Forest downstream; recommended versus random subsets	Descriptor evidence and data audit; Tables 5 and 6; Fig. 3
H2	The descriptor effect transfers between implementations	Graph-only and graph-plus-descriptor models in `chemistry-framework` and in Chemprop v2.3.1, on the same saved splits	Predictive performance; Tables 3 and 4; Fig. 2
H3	Route and role matter, but the best choice is endpoint-specific	Five role configurations; a harmonised routing experiment with Spearman, mutual-information and random controls; per-split paired comparisons	Descriptor routing and predictive roles; Tables 7 and 8; Fig. 4

Table 3. Regression performance (RMSE, mean ± SD over ten saved splits). Bold marks the best mean in each column; ties at the displayed precision are both bold.

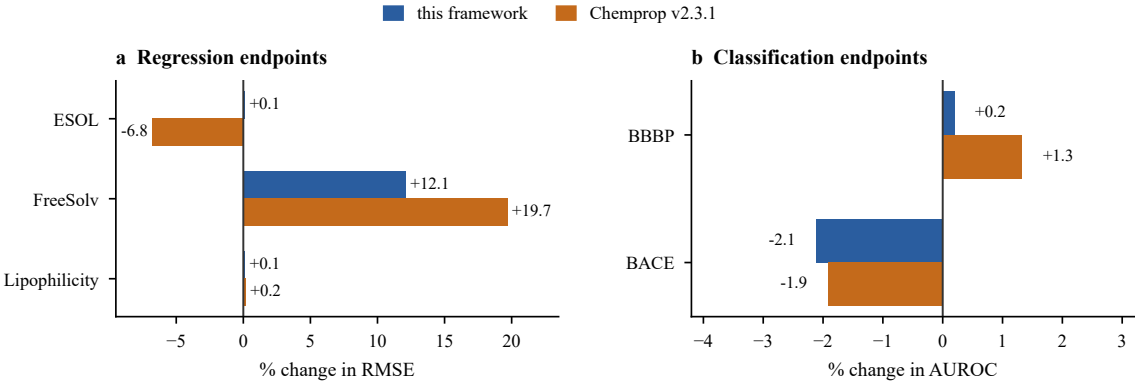
Model	ESOL	FreeSolv	Lipophilicity
Reference models			
Dummy	1.894 ± 0.197	3.619 ± 0.388	1.148 ± 0.039
ECFP4 + Random Forest	1.352 ± 0.093	1.795 ± 0.306	0.853 ± 0.028
RDKit 2D + Random Forest	0.637 ± 0.061	1.089 ± 0.175	0.748 ± 0.025
Descriptor MLP	$\mathbf{0.623 \pm 0.040}$	1.290 ± 0.446	0.683 ± 0.023
Chemprop v2.3.1			
Graph-only D-MPNN	0.791 ± 0.063	1.178 ± 0.320	0.641 ± 0.085
D-MPNN + descriptors (x_d)	0.844 ± 0.640	$\mathbf{0.946 \pm 0.261}$	0.639 ± 0.068
Framework			
Graph only (`residual_only`)	0.623 ± 0.092	1.251 ± 0.153	0.607 ± 0.031
Graph + descriptors (`direct_hybrid`)	$\mathbf{0.623 \pm 0.054}$	1.099 ± 0.249	$\mathbf{0.606 \pm 0.019}$
Prior + graph residual (`prior_residual`)	0.681 ± 0.113	1.175 ± 0.207	0.609 ± 0.036

D-MPNN with descriptors was best on FreeSolv (0.946) and BBBP (0.923); the RDKit 2D Random Forest remained best on BACE (0.822). After Holm correction only two paired comparisons excluded zero: the framework hybrid beat the RDKit 2D Random Forest on Lipophilicity ($\Delta = -0.142$, corrected 95% CI $[-0.179, -0.106]$, 10/10 splits) and trailed it on BACE (hybrid -0.037 AUROC, $[-0.058, -0.017]$, 0/10 wins); every other comparison was inconclusive, including the ESOL hybrid-versus-Random-Forest and hybrid-versus-graph-only contrasts (Additional file 1, Table S8). Descriptors outperformed ECFP4 fingerprints with Random Forests on every task, although the BACE margin was negligible (0.822 versus 0.822 at the displayed precision).

The framework and Chemprop graph only models were close: the framework was more accurate on ESOL (0.623 versus 0.791, nominal $p = 0.027$, Holm-adjusted $p = 0.053$ within the two external comparisons for that dataset), Chemprop was more accurate on FreeSolv (1.252 versus 1.178, not significant), and the remaining tasks were indistinguishable; across tasks the framework won between three and nine of ten splits (Additional file 1, Table S9). More informative than either ranking was the effect of adding descriptors (Fig. 2). In the framework the change in the mean was small on ESOL (+0.1%), Lipophilicity (+0.1%) and BBBP (+0.2%), clearly positive on FreeSolv (+12.1%), and slightly negative on BACE (−2.1%). Chemprop showed the same pattern on FreeSolv (+19.7%), Lipophilicity (+0.2%), BBBP (+1.3%) and BACE (−1.9%), but on ESOL its descriptor hybrid was better on nine of the ten splits and failed badly on one (test RMSE 2.76), so the mean fell by 6.8% with an SD of 0.64. That failure is diagnosed in Section : retraining the same configuration on that split with six seeds spans 0.75 to 6.12 RMSE, so the mean over one seed per split is not an estimate of a stable effect and its sign on ESOL is not interpreted. Reporting only that mean would have reversed the sign of the effect on ESOL, and the per-split record kept by the workflow is what makes the instability visible. On the four endpoints without such a failure the two implementations agreed on the sign and approximate size of the descriptor effect, which suggests that the effect is a property of the task and protocol rather than of the code.

Table 4. Classification performance (AUROC, mean ± SD over ten saved splits). Bold marks the best mean in each column; ties at the displayed precision are both bold.

Model	BBBP	BACE
Reference models		
Dummy	0.500 ± 0.000	0.500 ± 0.000
ECFP4 + Random Forest	0.905 ± 0.022	0.822 ± 0.028
RDKit 2D + Random Forest	0.912 ± 0.019	**0.822 ± 0.027**
Descriptor MLP	0.915 ± 0.028	0.787 ± 0.034
Chemprop v2.3.1		
Graph-only D-MPNN	0.911 ± 0.024	0.808 ± 0.056
D-MPNN + descriptors (x_d)	**0.923 ± 0.019**	0.793 ± 0.040
Framework		
Graph only (`residual_only`)	0.914 ± 0.034	0.802 ± 0.038
Graph + descriptors (`direct_hybrid`)	0.916 ± 0.029	0.785 ± 0.033
Prior + graph residual (`prior_residual`)	0.910 ± 0.034	0.783 ± 0.062



Positive values mean that adding the descriptors helped. ESOL (Chemprop): the negative mean comes from one split of ten, on which the descriptor hybrid reached a test RMSE of 2.76 while beating graph-only on the other nine. All values are computed from the saved per-split records.

Fig. 2. Effect of adding 210 RDKit descriptors to a graph-only D-MPNN, in two independently written implementations. Panel (a) shows regression endpoints and panel (b) classification endpoints, so the two metric families are not mixed on one axis. Bars are the change in the ten-split mean of the primary metric (RMSE in (a), AUROC in (b)), signed so that positive values mean lower error or higher AUROC. The ESOL value for Chemprop is dominated by one split, in which its descriptor hybrid reached a test RMSE of 2.76 while being better than graph-only on the other nine; retraining that configuration on that split with six seeds spans 0.75 to 6.12 RMSE, and bounding the descriptor inputs collapses the six retrainings to 0.65–0.72, so the ESOL bar is not an estimate of a stable effect. Computed from Tables 3 and 4.

Descriptor evidence shows what the data support, not what is universally important

A chemist approaching a new endpoint asks two things before any model is trained: are the records trustworthy, and which of the several hundred descriptors are actually supported by the training molecules? The audit answers the first, the scorecard and the selectors answer the second, and the two answers are independent of the learner. The audit found duplicated structures in two of the five benchmarks (Table 6, discussed below).

Figure 3 follows a single molecule through both checks, so that the numbers in the tables can be read on a concrete case.

With a Random Forest as learner (ESOL, FreeSolv, BBBP), recommended subsets had better mean scores than random subsets of the same size at every budget, and the corrected intervals excluded zero at all budgets on ESOL, at $K \leq 20$ on FreeSolv, and at $K = 5$ on BBBP; the margin shrank as K grew. At $K = 50$ on ESOL, 10 and 20 on FreeSolv, and the three larger budgets on BBBP, the recommendation still had the better mean but the corrected interval did not separate it from random draws. Against the full 210-descriptor pool the picture reverses: at $K = 5$ and $K = 10$ on ESOL the recommended subset was significantly worse, and every other budget was inconclusive (Additional file 1, Table S4).

A ten-draw control carries its own draw noise, so the comparison was repeated with 50 draws per split and budget

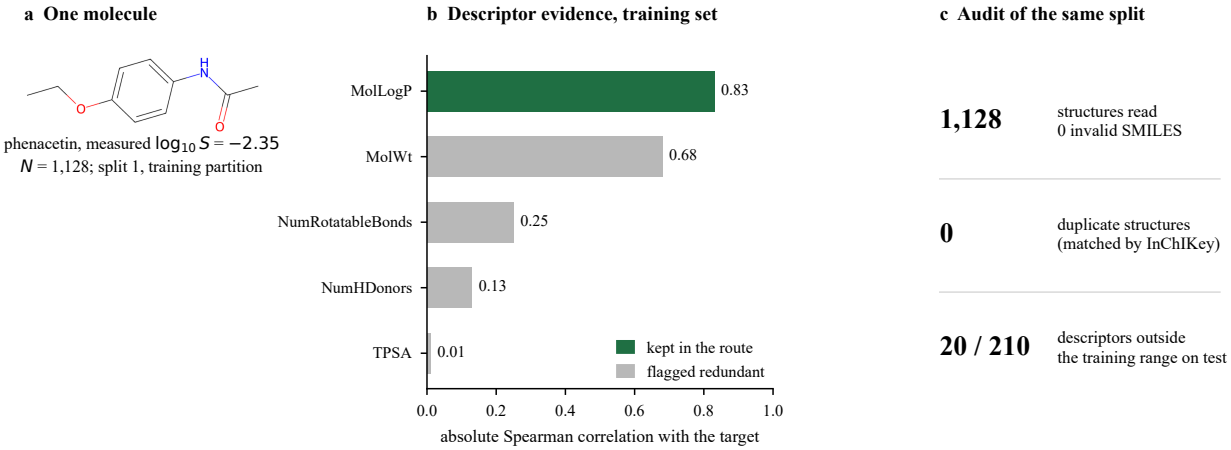


Fig. 3. One molecule followed through the audit and the descriptor scorecard. (a) Phenacetin in the ESOL dataset; its measured aqueous solubility is $\log_{10} S = -2.35$ and it belongs to the training partition of split 1. (b) Training-partition evidence for five of the 210 descriptors. Bars are absolute Spearman correlations with the measured solubility; a descriptor is grey if the recommender did not keep it, which in these cases is because its Pearson correlation with a higher-ranked descriptor exceeded 0.85. (c) Audit output for the same split. The full scorecard covers every descriptor and every split.

in all five endpoints, keeping every draw and testing the recommendation's rank inside the resulting distribution (Additional file 1, Table S5). The recommended subset had the better mean in all 20 dataset–budget cells, and the Fisher-combined rank test was significant in 16: all twelve regression cells ($p \leq 1.1 \times 10^{-3}$; mean percentile 0.88–1.00) and four of the eight classification cells (BBBP at $K = 5, 10, 20$ and BACE at $K = 5$, $p = 4.4 \times 10^{-4}$ to 0.036). The four that did not separate are the larger classification budgets (BBBP $K = 50$ and BACE $K = 10, 20, 50$), where a Random Forest on any 50 descriptors already approaches the full-pool AUROC: on BBBP the 50-draw mean is 0.774 at $K = 5$ but 0.905 at $K = 50$, against 0.916 for the full pool. The advantage of the evidence-based subset is therefore systematic, and largest where descriptors are scarce; it is not a claim that 50 chosen descriptors beat 50 random ones on a classification endpoint.

Across six selectors on five datasets (Table 5), Random Forest importance and mutual information ranked best on average, with mean ranks of 2.20 and 2.35, respectively. The framework recommender ranked fourth (3.65), was best in one of the 20 dataset-budget cells, and had mean pairwise Jaccard stability of 0.76. This stability was below Spearman ranking (0.88) and mutual information (0.80), but above Random Forest importance (0.65), MI–Pearson (0.57), and L1 (0.47). A Friedman test over the 20 dataset-budget cells rejects equality of the six selectors ($\chi^2 = 38.3$, $p = 3.3 \times 10^{-7}$), but after Nemenyi correction at $\alpha = 0.05$ (critical difference 1.69 ranks) the recommender is not separated from any individual selector, including the best: its gap to Random Forest importance is 1.45 ranks. The individual selector comparisons that survive Holm correction within a dataset are listed in Additional file 1, Table S7. These results support a transparent and reasonably stable candidate-generation heuristic, not a selector that consistently outperforms established alternatives. The selector match with the downstream Random Forest may favour RF importance. Read against the three criteria stated for H1, the 50-draw control meets (i) in all 20 cells on the mean comparison and in 16 on the rank test, the selector comparison fails (ii), and the stability analysis meets (iii).

The practical reading for a chemist is therefore: the scorecard tells you what the data can support, the recommender proposes a compact and stable subset, and the ranking among selectors is close enough that the choice should be made on inspectability and stability rather than on a small score difference.

The data audit finds problems that the training score cannot see

The audit found duplicated structures in two of the five benchmarks (Table 6). BBBP contained 64 structures recorded more than once (132 records, 6.5%), 11 of them with conflicting labels; ESOL contained 11 duplicated structures whose label spans stayed below the conflict threshold (largest span 1.03 log units, compared with a threshold of

Table 5. Six feature selectors on five datasets with a Random Forest downstream. Mean rank (1 = best) per dataset over $K \in \{5, 10, 20, 50\}$ and overall over 20 dataset-budget cells; Best denotes cells in which the selector had the best mean score; Jaccard: mean pairwise stability over splits and cells. Full results in Additional file 1, Table S6.

	Mean rank per dataset							
Selector	ESOL	FreeSolv	Lipo.	BBBP	BACE	Overall	Best	Jaccard
RF impurity importance	1.0	1.8	1.8	4.8	1.8	2.20	10	0.65
Mutual information	2.8	2.5	1.5	2.2	2.8	2.35	3	0.80
MI-Pearson	3.8	4.2	3.5	1.5	2.8	3.15	4	0.57
Framework recommender	3.8	3.2	3.5	3.8	4.0	3.65	1	0.76
Spearman top-K	3.8	3.2	5.0	5.8	5.8	4.70	0	0.88
L1 (lasso / L1-logistic)	6.0	6.0	5.8	3.0	4.0	4.95	2	0.47

Table 6. Structure-level audit by InChIKey. Duplicated structures are structures recorded more than once (records in parentheses). Conflicts are duplicated structures whose labels disagree (classification) or span more than half the target standard deviation (regression).

Dataset	Records	Unique InChIKeys	Duplicated structures	Conflicts	Largest span
ESOL	1,128	1,117	11 (22)	0	1.03
FreeSolv	642	642	0 (0)	0	0.00
Lipophilicity	4,200	4,200	0 (0)	0	0.00
BBBP	2,039	1,971	64 (132)	11	–
BACE	1,513	1,513	0 (0)	0	–

1.0478). Across the ten scaffold splits, the mean number of out-of-range descriptors was 19.4 ± 12.8 for ESOL, 20.8 ± 14.8 for Lipophilicity, and 15.1 ± 9.5 for BBBP; the corresponding fractions of test molecules affected were $6.3 \pm 2.5\%$, $3.7 \pm 0.7\%$, and $3.8 \pm 2.1\%$, respectively. These diagnostics are range-based indicators rather than a comprehensive measure of distribution shift. In BBBP the relationship was not monotone: the hybrid's weakest split (split 8) had 19 out-of-range descriptors and 8.8% of test molecules affected, but split 3 had almost as many (18) and gave the hybrid its best score (Additional file 1, Table S3). A range-based count is therefore a warning flag, not a predictor of which split will be hardest.

The duplicate-controlled BBBP sensitivity analysis retained 1,960 unique structures after excluding the 11 conflicting label groups and retaining one record for each of the 53 remaining duplicated groups. On ten new scaffold manifests, mean AUROC was 0.8656 ± 0.0577 for the framework graph only model and 0.8637 ± 0.0422 for its descriptor hybrid. Chemprop v2.3.1 achieved 0.8686 ± 0.0398 with graph only and 0.8744 ± 0.0428 with the 210 molecular descriptors, while RDKit 2D plus Random Forest achieved 0.8940 ± 0.0305 (Additional file 1, Table S16). The hybrid minus graph mean difference was -0.0019 for the framework (5/10 splits favoured the hybrid) and $+0.0057$ for Chemprop (9/10 splits favoured the hybrid). For the framework the paired 95% t interval included zero, so these results do not establish a conclusive advantage from descriptor augmentation in this workflow. The Chemprop hybrid-minus-graph difference was in the same direction but smaller, and its interval also included zero; after Holm correction within the five sensitivity comparisons neither implementation separates (Additional file 1, Table S17). The exact ranking differed from the primary benchmark. Scores were lower for the learned models on the sensitivity data, but this change cannot be attributed to duplicate leakage alone because both data composition and split manifests differed.

Descriptor routing depends on feature budget and endpoint

The harmonised routing results show that the value of a compact descriptor route depends on the endpoint (Table 7). The full 210-descriptor hybrid had the lowest mean RMSE on ESOL (0.614 ± 0.039) and FreeSolv (1.038 ± 0.214). On Lipophilicity the recommender route had the lowest mean of the six routes (0.581 ± 0.041), below the full descriptor set (0.606); on BACE the graph-only model had the highest mean AUROC (0.798), and the six routes spanned only 0.014, less than the between-split SD. On BBBP every route lay within 0.007 AUROC of the best. The random-subset control is the important comparator. The recommender beat it on Lipophilicity (0.581 versus

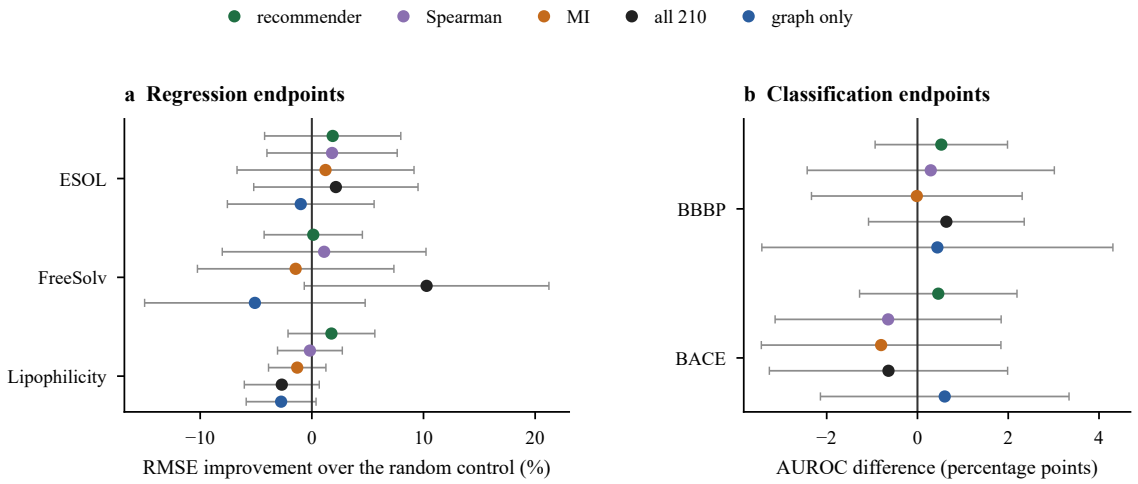


Fig. 4. Compact descriptor routes compared with the matched random control. Points are mean paired differences over ten splits and error bars are population SDs. Positive values mean that the route had a lower RMSE (a) or a higher AUROC (b) than a random subset of the same size drawn at the same split-specific budget K^*. The largest mean improvement is the full descriptor set on FreeSolv (about 10%), but every interval includes zero: at this sample size no route is separated from its control.

Table 7. Harmonised evidence-guided descriptor routing. Entries are test RMSE for ESOL, FreeSolv, and Lipophilicity, and test AUROC for BBBP and BACE; values are mean ± population SD over ten saved splits. All route checkpoints are selected by minimum validation loss, and the recommended route's K^* is selected using the primary validation metric at that checkpoint. The random route averages ten random-subset draws per split. Bold text marks the best mean in each row. The two univariate selectors (Spearman and mutual information) are reported in Additional file 1, Table S10, together with the full checkpoint-level audit of 250 model runs (Table S18).

Dataset	Metric	Graph only	All 210	Recommender	Random
ESOL	RMSE ↓	0.638 ± 0.074	**0.614 ± 0.039**	0.620 ± 0.078	0.631 ± 0.057
FreeSolv	RMSE ↓	1.205 ± 0.164	**1.038 ± 0.214**	1.147 ± 0.140	1.149 ± 0.141
Lipophilicity	RMSE ↓	0.607 ± 0.031	0.606 ± 0.019	**0.581 ± 0.041**	0.591 ± 0.027
BBBP	AUROC ↑	0.914 ± 0.034	**0.916 ± 0.029**	0.915 ± 0.023	0.910 ± 0.020
BACE	AUROC ↑	**0.798 ± 0.038**	0.786 ± 0.033	0.797 ± 0.036	0.792 ± 0.042

0.591), but was within one SD of it on ESOL (0.620 versus 0.631), FreeSolv (1.147 versus 1.149), BBBP (0.915 versus 0.910) and BACE (0.797 versus 0.792), and every paired recommender-minus-control difference had a Holm-corrected interval that included zero (Additional file 1, Table S11). The same correction left the Lipophilicity gap to the full descriptor set inconclusive, so the apparent benefit of compactness is not established at this sample size either. A compact route therefore does not appear to require an informed selection rule on these endpoints; what matters is that it is tested rather than assumed. Per-split outputs and checkpoint decisions are archived in the repository; the checkpoint selection rule and result-file inventory are documented in Additional file 1, Table S18.

Configurable predictive roles are useful but not uniformly beneficial

The best role assignment was also task dependent (Table 8). The descriptor-only role was best on ESOL (0.619), direct fusion on FreeSolv, Lipophilicity, and BBBP, and graph only on BACE (0.802 versus 0.785 for direct fusion). The margins were often small: on Lipophilicity and ESOL the four learned roles other than the prior and prior-plus-residual variants spanned 0.004 and 0.018 RMSE respectively, so the ordering should be read as a trend across ten splits rather than as a separated ranking. No paired interval excluded zero for the prior/residual model against graph only on any task, and on ESOL it had the largest split-to-split variability of the learned models. At the level of training seeds, variation was small relative to split variation for most models (SD over three seeds 0.003 - 0.007 AUROC on BBBP), but reached 0.013 for the prior/residual model, about a third of its between-split SD of 0.034 (Additional file 1, Table S12).

Table 8. Predictive-role ablation (primary metric, mean ± SD over ten splits; RMSE for ESOL, FreeSolv, and Lipophilicity, AUROC for BBBP and BACE). Bold text marks the best mean per column.

Configuration	ESOL	FreeSolv	Lipophilicity	BBBP	BACE
`prior_only`	0.913 ± 0.073	2.015 ± 0.116	1.009 ± 0.036	0.812 ± 0.030	0.660 ± 0.055
`direct_only`	$\mathbf{0.619 \pm 0.045}$	1.350 ± 0.458	0.683 ± 0.034	0.903 ± 0.041	0.785 ± 0.046
`residual_only`	0.623 ± 0.092	1.251 ± 0.153	0.607 ± 0.031	0.914 ± 0.034	$\mathbf{0.802 \pm 0.038}$
`direct_hybrid`	0.623 ± 0.054	$\mathbf{1.099 \pm 0.249}$	$\mathbf{0.606 \pm 0.019}$	$\mathbf{0.916 \pm 0.029}$	0.785 ± 0.033
`prior_residual`	0.681 ± 0.113	1.175 ± 0.207	0.609 ± 0.036	0.910 ± 0.034	0.783 ± 0.062

Robustness of the inferences

Four checks asked whether the conclusions survive a change of statistical view, a change of control protocol, a change of two conventions, and a change of environment.

The primary interval is a Nadeau–Bengio corrected resampled t interval, which is conservative when training sets overlap. Repeating all 184 paired comparisons with an exact sign-flip permutation test, which is computationally exact but makes no overlap correction and is therefore more permissive, reached the same decision in 175 of 184 comparisons and found 23 significant differences rather than 14. The nine disagreements all involve effects whose intervals barely include or exclude zero; none changes a direction, so the conclusions are unchanged when the stricter corrected-interval view is used (Additional file 1, Table S19).

For the routing experiment, none of the 15 route-versus-control comparisons reached significance under the permutation test either (Holm-adjusted $p = 1.000$ throughout), so the finding that compact routes are not reliably better than random subsets of the same size does not depend on the interval used.

Two conventions were also varied rather than assumed. Changing the redundancy threshold of the recommender over 0.70–0.90 moved the selected set substantially (mean Jaccard 0.51–0.77 against the default) but changed its downstream score by at most 0.024 RMSE on ESOL, 0.015 on FreeSolv and 0.005 AUROC on BBBP; changing the encoder width over 64–256 kept the sign of the descriptor effect on FreeSolv and changed its size between 7% and 19% (Additional file 1, Tables S20 and S21). In both cases the convention was not what the conclusion depended on.

Finally, because the framework and the Chemprop comparator ran in separate environments with different RDKit releases, the 210-column descriptor block was exported from both and compared cell by cell: 99.87% of 1,999,620 cells agree. The differences are confined to ten descriptors, mostly functional-group counts, and are the reason the cross-implementation comparison is described as a replication rather than as an identical-input ablation.

The divergent run, diagnosed

Two runs in the study departed from every other run of the same configuration, both on ESOL split 3: Chemprop's descriptor hybrid reached a test RMSE of 2.76 while being better than graph-only on the other nine splits, and the framework's prior/residual route reached 13.23. Excluding them would have been a post-hoc decision, so both were investigated instead.

Retraining the framework route with three independent training seeds reproduced the failure in all three runs (13.24, 13.25, 13.22), whereas the direct hybrid on the same split, descriptors and budget gave 0.707, 0.693 and 0.727. The failure is therefore a property of the route and the split, not of one optimisation trajectory. The mechanism is visible in the predicted values: the ridge prior, fitted on the same 50 descriptors, produced test predictions between -8.09 and 128.72 for targets between -8.40 and 1.58, and the residual head adds only a small correction on top of that.

The cause is a single descriptor. `Ipc` (information content) has a training mean of 2.33×10^5 and a training SD of 1.70×10^6 on this split, and the split-3 test set contains a molecule with $\mathrm{Ipc} = 5.29 \times 10^9$, which is 3114 training SDs from the mean. The harness standardises descriptors and replaces infinities by ± 3, but it does not bound large finite values, so the network and the ridge prior receive an input of 3113.7; the largest standardised test value on the

other nine ESOL splits is 30.0. The nonlinear hybrid encoder absorbs this because its descriptor branch is normalised after the first linear layer; the prior/residual route does not, because it adds a linear prediction directly.

Chemprop's hybrid is less stable on the same split. Retraining its archived configuration with six seeds, holding the split, the descriptor block, the model and every hyperparameter fixed, gives test RMSE 0.75, 1.01, 1.20, 2.33, 5.40 and 6.12, with maximum absolute predictions of 8.2, 9.7, 12.2, 24.7, 50.2 and 57.2 against targets spanning -8.40 to 1.58. The archived value of 2.76 is therefore one draw from an unstable configuration rather than a unique accident: three of the six retrained runs exceed 2.0 and three stay below 1.3. Because the archived study trained one seed per split, the Chemprop ESOL descriptor effect of -6.8% is a property of the seed used on split 3, and we do not interpret its sign; the within-system framework contrast for the same endpoint, which is stable across splits, puts the effect at approximately zero.

The intervention confirms the mechanism. Winsorising the standardised descriptor block at ± 5 training standard deviations, with the data, split, model, seeds and all other hyperparameters unchanged, gives 0.666, 0.699, 0.719, 0.698, 0.653 and 0.677 RMSE for the same six seeds: a spread of 0.066 instead of 5.37, and a mean of 0.685 against 2.802 without the bound. Bounding the magnitude of the descriptor inputs, and nothing else, therefore removes both the tail and the instability. The stabilised hybrid is also close to the graph-only arm on the same split (0.673), so once the input is bounded the ESOL descriptor effect is approximately zero rather than negative; the framework's within-system contrast reaches the same conclusion ($+0.1\%$). The failure is a property of the input magnitude, not of Chemprop's training procedure.

The range audit had already flagged the split: ESOL split 3 has 44 of 210 descriptors outside the training range and 11.5% of test molecules affected, more than any other ESOL split. The aggregate count was not sufficient, however, because it does not separate a descriptor one unit beyond the training range from one 3114 standard deviations beyond it, and Lipophilicity split 9 has more out-of-range descriptors (60) with no divergence. Magnitude separates them: the largest standardised test value on ESOL split 3 is 3114, against at most 30.0 on the other nine ESOL splits and at most 62 on the Lipophilicity splits; the only comparably large values in the audit are BBBP splits 6 and 8 (430 and 478, both on `Kappa3`), where nothing diverged. A maximum-magnitude check is therefore a warning to inspect rather than a verdict, and we report it in the audit output alongside the range count, with values in the hundreds of training standard deviations treated as a reason to winsorise the descriptor block before interpreting the score.

Discussion

The results support an evidence-guided approach to molecular property modelling, not a universal ranking of model families. Across five independent endpoints, descriptor-based, graph-based, and hybrid predictors each had tasks on which they performed well. Adding the 210 RDKit descriptors changed performance in the same direction in two independently written implementations on four of the five tasks, and by a similar amount: a clear gain on FreeSolv and a change of a few percent or less elsewhere. On ESOL the two disagreed: Chemprop's hybrid was better than graph-only on nine of ten splits and collapsed on the tenth, and retraining that configuration on that split spans 0.75 to 6.12 RMSE across six seeds, so the disagreement is about the stability of one configuration rather than about the endpoint. Both failures are removed by bounding the same descriptor block, and the extreme value the audit reports is an `Ipc` value 3114 training standard deviations outside the training range; both are visible only because the workflow keeps per-split and per-run records rather than a summary.

It is worth stating what this means for the chemist who has one dataset in hand. The default "use the graph" or "use all descriptors" habit is not wrong, but it is a guess, and the cost of testing it is now small. On phenacetin's endpoint the same graph encoder gave a test RMSE of 0.681 without descriptors and 0.595 with them on that split, a difference of 0.086; averaged over ten splits the graph-only and full-descriptor means are within 0.001 RMSE of each other on ESOL (0.623 versus 0.623), which is exactly why a single split should not be used as a decision rule. The same holds for compact routes: the recommender gave the lowest mean on Lipophilicity (0.581 versus 0.591 for random

subsets), although the paired difference did not survive correction, and it did not separate from the control on any other endpoint. None of these numbers is a universal rule, and the framework deliberately does not turn any of them into one. What it does is make the alternatives cheap enough to compare on the user's own data, with the same split and the same record.

Two different questions are entangled in a comparison between software packages, and they should be separated. The first is whether descriptors help a graph model at all; the unconfounded estimate of that is the within-system contrast, graph only versus the same encoder with descriptors, which requires no second implementation. The second is whether that effect is a property of the endpoint or an accident of one codebase; that is a replication question, and it is answered by an independently written implementation showing the same sign and roughly the same size. Our design answers the second question and supports the inference by the first: the within-system contrasts show the effect, and the agreement with Chemprop v2.3.1, which differs in encoder, featurisation, optimiser and training loop, shows that the effect survives a change of code. It is not an architecture-controlled decomposition, and we do not read it as one. The price of that reading is paid in the Methods: the descriptor blocks exported from the two environments agree in 99.87% of cells, so the two systems saw effectively, but not exactly, the same input.

The contribution this paper claims is correspondingly narrow and, we think, useful. It is not a new predictor or a new state-of-the-art selector. It is the combination of (i) an audit that turns data problems and descriptor evidence into inspectable artefacts before any model is fitted, (ii) a configuration layer in which the same split, checkpoint rule and evaluation produce the alternatives, and (iii) a ten-split record in which a divergent run cannot hide inside a mean. The audit found duplicates in two of five benchmarks, one of them with conflicting labels, and the scorecard showed that training-partition evidence can produce a compact candidate set—though on ESOL the two smallest subsets scored worse than the full 210-descriptor pool, which is the honest limit of the rule. What neither tool does is name the important descriptors: the recommender is a candidate-generation heuristic built on a fixed relevance-and-redundancy rule, it ranked fourth of six selectors, and a block test over the 20 dataset–budget cells, while rejecting equality of the selectors, does not separate it from any individual selector, including the best. Its value is that the proposal is compact, stable across splits (mean Jaccard 0.76) and explainable feature by feature.

For a reader who wants a rule rather than a framework, the work supports a short checklist with explicit thresholds. Before fitting: curate duplicated structures whose labels conflict, because a scaffold split can place the same structure on both sides; in these five benchmarks that affected BBBP (64 duplicated structures, 11 with conflicting labels) and, without label conflicts, ESOL. Before interpreting a score: check how many descriptors lie outside the training range for the test molecules (6.3% of test molecules on ESOL, 3.8% on BBBP) and, more importantly, the largest standardised value they produce, because a maximum in the hundreds or thousands of training standard deviations means the score describes extrapolation rather than validation; ESOL split 3 reaches 3114 and is the split on which two routes diverged. Before choosing a representation: run at least graph only, the full descriptor block and one compact block on the same splits, and report per-split values, because on these endpoints the best choice differed between tasks and the margins were often a few hundredths of a metric unit. These thresholds are conventions drawn from this study, not validated cut-offs, and a prospective application should re-derive them.

Several limitations remain. Ten split manifests still provide limited statistical power for differences of a few hundredths of a metric unit, and overlapping training sets make paired intervals descriptive rather than definitive; the permutation tests reported in Additional file 1 agree with the corrected intervals in 175 of 184 comparisons, but they do not correct for that overlap and are correspondingly more permissive. Hyperparameters were fixed a priori rather than tuned. Two one-factor sensitivity analyses are reported in Additional file 1: varying the encoder width over 64–256 leaves the sign of the descriptor effect unchanged and changes its size between roughly 7% and 19% (Table S21), and varying the redundancy threshold over 0.70–0.90 changes the selected descriptor set substantially (Jaccard 0.51–0.77 against the default) while the downstream score moves by at most 0.024 RMSE (Table S20). Settings not varied here, such as the size of the downstream forest or the learning rate, could change the size of the small effects; tuning was not performed, so the reported values characterise a fixed protocol rather than an optimal one. The descriptor blocks

were generated by different RDKit releases in the two environments, which affects 0.13% of descriptor cells. The primary blood-brain barrier benchmark retained duplicate structures, including groups with conflicting labels, and the audit found that tautomeric representations can produce different scaffold strings for the same InChIKey. The duplicate-controlled analysis addresses sensitivity to curation, but changes to data composition and splits prevent a causal attribution to leakage alone. The descriptor-only reference and the framework's direct descriptor model were produced by different runners. The experiments use two-dimensional inputs, five endpoints, and retrospective splits; they do not test prospective use, multi-task training, or three-dimensional and mixture inputs, for which the framework exposes interfaces but no evidence. A prospective benchmark in which the audit and the route choices are fixed before the test data are unblinded would be the natural next step, together with extensions to other endpoints and to duplicate-aware partitioning.

Conclusions

We stated three hypotheses in the Background so that they could fail. The results answer them as follows.

H1, evidence is usable: partly supported. Criterion (i) is met: with a Random Forest downstream, the recommended subset had the better mean in all 20 dataset–budget cells, and the 50-draw rank test separated it from random subsets in 16 of them, all twelve regression cells and four of the eight classification cells. Criterion (ii) is not met: the rule ranked fourth of six selectors overall (mean rank 3.65, against 2.20 for Random Forest importance), and the block test, while rejecting equality, does not separate it from the best selector. Criterion (iii) is met: the mean Jaccard overlap of the selected sets across splits was 0.76. The correlation with the target is a weak guide to which descriptors a model will actually use, so the honest description is a stable and inspectable way to generate a compact candidate set, not a claim that the selected descriptors are the important ones.

H2, the descriptor effect transfers: supported on four of five endpoints, and consistent on the fifth once diagnosed. Adding 210 descriptors to a graph model changed the error by a similar amount in two independently written implementations on FreeSolv ($+12.1\%$ and $+19.7\%$), Lipophilicity ($+0.1\%$ and $+0.2\%$), BBBP ($+0.2\%$ and $+1.3\%$) and BACE (-2.1% and -1.9%). ESOL was the exception, and it is instructive: Chemprop's descriptor hybrid was better than graph-only on nine of ten splits and failed on one (test RMSE 2.76). Retraining that configuration spans 0.75–6.12 across six seeds, and bounding the descriptor inputs collapses the spread to 0.65–0.72, close to the graph-only arm on the same split; the ESOL effect is therefore approximately zero rather than reversed, which is what the framework's own contrast gives ($+0.1\%$). A mean without its per-split record would have produced a confident and wrong statement about chemistry, and the same record is what made the failure diagnosable. The practical consequence is that a descriptor effect should be reported per split, and a descriptor block bounded, before either is interpreted.

H3, route and role matter but are endpoint-specific: supported. The best route differed by endpoint: the full 210-descriptor hybrid on ESOL and FreeSolv, the recommender on Lipophilicity, graph-only on BACE, and no distinguishable winner on BBBP. The best predictive role also changed with the endpoint. Crucially, no compact route separated from a random subset of the same size once paired uncertainty was included, so the route is worth testing but should not be sold as a source of large gains on these endpoints.

Taken together, the three answers point to a practical rule for a chemist: test the alternatives on your own splits, keep the per-split record, and treat a mean that hides one divergent run as unfinished. `chemistry-framework` exists to make that rule cheap to follow. An experiment is a plain-text configuration file; the five commands behind it read the data, audit it, validate the configuration, fit the declared representations, and write a record containing the configuration hash, split seed, predictions and checkpoint. A colleague or referee can therefore see which choice was made, on which evidence, and what happened when it was changed. The framework does not promise a universally better predictor, and the results here give no reason to expect one.

Supplementary information

Additional file 1: Supplementary results (PDF). Dataset overview, target statistics, and provenance (Table S1); fixed hyperparameters (Table S2); scorecard diagnostics and split-level shift analysis (Table S3); recommended versus random subsets and the full pool (Table S4); high-repetition random-subset control (Table S5); six-selector results, stability and budget curves (Tables S6 and S7, Figure S1); internal and external paired comparisons (Tables S8 and S9); harmonised descriptor-routing results and their paired comparisons (Tables S10 and S11); training-seed variance (Table S12); secondary metrics and per-split scores (Tables S13 and S14); advisor case study on Caco-2 (Table S15); duplicate-controlled BBBP sensitivity analysis and its paired comparisons (Tables S16 and S17); checkpoint protocol and result-file inventory (Table S18); distribution-free checks of the paired comparisons and the random-subset control (Table S19); sensitivity of the recommender to the redundancy threshold and of the descriptor effect to encoder width (Tables S20 and S21).

Abbreviations

AP, average precision; AUROC, area under the receiver operating characteristic curve; BACE-1, beta-secretase 1; BBBP, blood–brain barrier penetration; CI, confidence interval; D-MPNN, directed message-passing neural network; ECFP, extended-connectivity fingerprint; GIN, graph isomorphism network; GNN, graph neural network; InChI, International Chemical Identifier; InChIKey, International Chemical Identifier key; LARS, least angle regression; MAE, mean absolute error; MI, mutual information; MI-Pearson, selector using mutual-information relevance and Pearson-correlation redundancy; MLP, multilayer perceptron; QSPR, quantitative structure–property relationship; QSAR, quantitative structure–activity relationship; RF, Random Forest; RMSE, root mean squared error; SD, standard deviation; SMILES, simplified molecular input line entry system; TDC, Therapeutics Data Commons.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The datasets are publicly available through MoleculeNet [8]. Split manifests, configuration files, per-split results, the scripts that produced them, and the manuscript sources are available at `https://github.com/soratori1025/Framework-Chemistry` (release tag `v1.0-paper1`, archived for this submission). Documentation, including a quick-start guide and the configuration reference, is maintained in the same repository.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

D.K.N.N. conceived the framework, developed the software, performed the experiments, analysed the results, and wrote the manuscript.

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Methods

Framework and configuration

`chemistry-framework` organises an experiment as data loading, configuration validation, splitting, featurisation, feature analysis, model construction, training, and evaluation (Fig. 1). Molecules are supplied as strings in the simplified molecular input line entry system (SMILES). A YAML (human-readable data serialisation) configuration declares the input file and targets, named feature blocks, routes that assign blocks to tasks, priors, task types, split strategy, and model, training, evaluation, and analysis settings; it is parsed against a typed schema before any computation. Feature definitions and resolved feature names are hashed, so cached matrices and results can be traced to the definitions that produced them. Feature blocks include element counts, bond-pair counts, substructure counts from SMILES arbitrary target specification (SMARTS) patterns, Morgan fingerprints [3], and the RDKit 2D descriptor set (210 descriptors in RDKit 2023.09.6) [61]; model components, implemented in PyTorch [62], include directed message-passing neural network (D-MPNN) and graph isomorphism network (GIN) encoders, a multilayer perceptron (MLP) descriptor encoder, fusion modules, and task-specific heads. A command-line interface (CLI) and a Python application programming interface (API) expose the commands `inspect`, `validate`, `prepare`, `train`, and `predict`. Installation, the configuration reference, and descriptions of every output file are maintained with the source code (`docs/` and `README.md` in the repository); the five commands above are also wrapped by a single reproduction script that reruns the benchmark suite. Interfaces for 3D, reaction, mixture, and protein inputs exist but are not evaluated here. The test suite (61 tests) covers configuration parsing, featurisation, feature analysis, the dataset advisor, and training.

Feature evidence and dataset audit

Training-derived scorecard statistics are computed on the training partition. For each feature the scorecard records provenance, including domain, source and definition, training support (number of molecules with a non-zero value), training mean, standard deviation and range, the Spearman correlation with each target, collinear partners ($|r| \geq 0.95$ by default) and, when supplied, bootstrap evidence for prior coefficients and model-dependent group contributions such as permutation or group-removal effects. An ordered rule assigns each feature an evidence class that can be, *out of distribution/unsupported* (zero support or held-out values outside the training range), *weak/uncertain*, *redundant*, *supported* (a sign-consistent prior coefficient whose bootstrap 95% interval excludes zero, together with a positive model contribution in at least 80% of repeats) or *model-dependent*. Because the out of distribution class uses held-out ranges, it serves post hoc diagnosis only and is never used for selection; it is a simple range based applicability-domain check [30, 31].

The scorecard and recommender serve complementary but distinct purposes. The scorecard is an evidence-documentation and diagnostic layer: it records what is known about a feature under the current dataset and split, but does not itself select features or decide how a feature block enters a model. The recommender is an optional candidate-generation heuristic that returns a compact subset under a fixed relevance and redundancy rule. In particular, the recommender does not consume the scorecard's categorical evidence classes, learn endpoint-specific weights, or

automatically alter model architecture. The user remains responsible for deciding whether and how a proposed subset is used.

The dataset advisor (`inspect`) reports SMILES validity, repeated SMILES strings, scaffold diversity [63], Tanimoto diversity of Morgan fingerprints [3, 64], target skewness, outliers, and class balance, and drafts a starting configuration. For this study, an audit script in the repository additionally matched structures by the International Chemical Identifier key (InChIKey) [65]; a duplicated structure was counted as a conflict if its labels disagreed (classification) or spanned more than half the target standard deviation (regression). Covariate shift between partitions was summarised in two ways: as the number of descriptors with any test value outside the training range and the fraction of test molecules affected, and as the largest standardised test value, obtained by fitting the descriptor standardisation on the training partition and recording the maximum absolute value over the test partition (with the responsible descriptor), together with the number of test molecules exceeding ten training standard deviations. The second summary is reported because a range count treats a value one unit outside the training range and a value thousands of standard deviations outside it identically, and the two have very different consequences for a standardised model input. We report both illustrative split-level diagnostics and the mean and population standard deviation over the ten saved scaffold splits where available. For BBBP, the audit also identified duplicate InChIKeys crossing partition boundaries in several scaffold manifests. Inspection of the affected records showed tautomeric representations that produced different Murcko scaffold strings, so scaffold grouping alone did not guarantee structure identity across partitions.

Predictive roles

A route maps named feature blocks to a task, so the same block can be reassigned without changing the feature calculator or training loop. In the *direct* role, features are encoded by an MLP and optionally concatenated with the graph representation; in the *prior* role, a feature set x_p defines a baseline fitted on the training partition; in the *residual* role, a network corrects that baseline:

$$\hat{z}(\mathscr{G}, x_d, x_p) = h_\theta([\,\phi_\theta(\mathscr{G})\,;\,\psi_\theta(x_d)\,]) + f_p(x_p), \tag{1}$$

where ϕ_θ encodes the molecular graph $\mathscr{G}$, ψ_θ encodes directly routed descriptors x_d, h_θ is the readout and output head, and either encoder can be disabled. For regression $\hat{y} = \hat{z}$; for binary classification $\hat{z}$ is a logit and $\hat{y} = \sigma(\hat{z})$. The pattern follows Δ-machine learning [50], with a statistical baseline of physicochemical descriptors in place of a lower-level calculation. The framework fits priors by ridge, least-squares, or logistic regression, optionally with fixed coefficients; in the benchmark, priors were fitted with scikit-learn [66] (ridge, $\alpha = 1$; logistic, $C = 1$, using the log-odds from `decision_function`) on six descriptors (`MolWt`, `MolLogP`, `TPSA`, `NumHDonors`, `NumHAcceptors`, `NumRotatableBonds`), frozen, and supplied as an input. The network was trained on the original target (Huber loss [67] or binary cross-entropy on logits). Listing 1 shows how roles are declared.

Feature recommendation and comparison selectors

The framework recommendation at budget K (`select_evidence_recommendation`) uses the training partition only: (1) compute the absolute Spearman correlation s_j of each feature with the target, setting $s_j = 0$ for features with standard deviation below 10^{-6}; (2) sort features by s_j; (3) add a feature if $s_j \geq 10^{-4}$ and its absolute Pearson correlation with every selected feature is at most 0.85; (4) if fewer than K features pass, pad with the next-ranked features. Padding was never triggered: at least 128 descriptors passed step 3 in every split. The rule is a transparent relevance and redundancy heuristic related to correlation based and minimum redundancy selection [40, 41]; it uses the scorecard's association and collinearity statistics but not its evidence classes. It is therefore presented as a reproducible candidate-generation rule for this workflow, not as a novel feature-selection algorithm or a claim that its selected descriptors are universally important.

Five selectors served as comparators at $K \in \{5, 10, 20, 50\}$: Spearman top-K; mutual information (MI) top-K with the

Listing 1. Configuration excerpt showing the three predictive roles in Eq. (1). The excerpt is validated against the framework schema. Removing the task prior gives a direct hybrid, disabling the descriptor branch gives a prior and residual model, and setting the graph encoder to none gives a descriptor-only model.

```
features:
  physchem: # six descriptors used by the prior
    type: rdkit_descriptors
    names: [MolWt, MolLogP, TPSA, NumHDonors,
            NumHAcceptors, NumRotatableBonds]
  rdkit2d: # full RDKit 2D descriptor pool
    type: rdkit_descriptors
    families: [all]
priors:
  logS_prior: {use: [physchem], fit: ridge} # prior role
routes:
  logS: [rdkit2d] # direct role
tasks:
  logS: {target: logS, kind: scalar, route: logS, prior: logS_prior}
split: {method: scaffold, fractions: [0.8, 0.1, 0.1], seed: 1}
model:
  structure: {encoder: dmpnn, hidden_dim: 128, depth: 3} # graph branch
  descriptors: {enabled: true}
  fusion: concat
  prior: frozen
```

k-nearest-neighbour estimators of scikit-learn ($k = 3$) [38, 39]; an MI-Pearson selector that greedily maximises mutual information relevance while penalising mean absolute Pearson correlation with selected features; Random Forest impurity importance (100 trees, depth ≤ 12) [44]; and L1 selection, where L1 denotes absolute-value regularisation (order of entry on the least angle regression (LARS) lasso path for regression, or absolute coefficients of an L1-penalised logistic regression with $C = 1$ for classification) [42, 43]. L1 selection used a standard scaler fitted on the training partition only. Random subsets (ten draws per split) and the full pool were additional references on ESOL, FreeSolv, and BBBP. Selected subsets were evaluated with a Random Forest (100 trees). Stability was the mean pairwise Jaccard index of the selected sets over the ten splits [47, 48].

Evidence-guided descriptor routing

To test whether feature evidence can guide model configuration, the 210-descriptor input of the direct hybrid was replaced by a compact route. For each split, the recommended subset was evaluated at $K \in \{10, 20, 50\}$, and the budget with the best primary validation metric (K^*) was retained. Spearman, mutual information (MI), and random subsets reused the same split-specific K^* as matched controls; the random route averaged ten random-subset draws per split. An auxiliary prior and residual model also used the recommended K^* subset as its prior input (`prior_rec`); because this changes the predictive role as well as the descriptor route, it is retained in the dataset JSON outputs rather than in the matched direct-hybrid selector table. All compact-route checkpoints were selected by minimum validation loss, and K^* was selected using the primary validation metric evaluated at that checkpoint. The graph-only and full-descriptor references used the same validation-loss checkpoint rule. Thus, checkpoint selection is harmonised across all five datasets and all routing comparisons. The full checkpoint-level audit covers 250 model runs and is recorded in `benchmarks/results/routing_checkpoint_audit.csv`; dataset-specific result files are listed in Additional file 1, Table S18.

Datasets and splits

Five datasets from MoleculeNet were used (Table 9): the ESOL aqueous-solubility benchmark [69], FreeSolv hydration free energy [70], Lipophilicity [71], blood-brain barrier penetration (BBBP) [72], and the beta-site amyloid precursor protein cleaving enzyme 1 (BACE-1) inhibition dataset, referred to as BACE [73], in their DeepChem distributions; SMILES that RDKit could not parse were removed (11 in BBBP), and no deduplication was applied

Listing 2. Scaffold partitioning as implemented in `benchmarks/prepare_moleculenet_data.py`. The same seed reproduces the manifest byte for byte; the manifest file, not this code, is the object every model is trained on.

```
for seed in 1..10:
    groups <- molecules grouped by Bemis-Murcko scaffold (acyclic -> singleton)
    groups <- shuffle(groups, rng(seed))
    groups <- sort(groups, key = size, descending)
    target <- (0.8, 0.1, 0.1) * N
    counts <- (0, 0, 0)
    for g in groups:
        part <- argmin_k( counts[k] / target[k] )
        assign(g, part); counts[part] += size(g)
    save_manifest(seed, train, valid, test)
```

Table 9. MoleculeNet datasets, prediction endpoints, primary metrics, and saved split manifests. Counts refer to the processed datasets used in the experiments.

Dataset	Endpoint	Task	Metric	N	Train	Valid.	Test	Split
ESOL	Aqueous solubility ($\log_{10}$ mol L^{-1})	Regression	RMSE	1,128	902	113	113	Scaffold
FreeSolv	Hydration free energy (kcal mol^{-1})	Regression	RMSE	642	513	64	65	Random
Lipophilicity	Octanol/water $\log D$ at pH 7.4	Regression	RMSE	4,200	3,360	420	420	Scaffold
BBBP	Blood-brain barrier penetration	Binary class.	AUROC	2,039	1,631	204	204	Scaffold
BACE	BACE-1 inhibition	Binary class.	AUROC	1,513	1,209	152	152	Scaffold

in the primary benchmark. Structures were additionally matched by InChIKey to audit duplicate allocation across the saved partitions. For BBBP, we performed a sensitivity analysis after removing the groups with conflicting labels and retaining one record for each remaining InChIKey. New scaffold split manifests were generated and reused across every model in that sensitivity comparison. Each primary dataset has ten saved 80/10/10 manifests reused by every model. Scaffold splits group molecules by Bemis-Murcko scaffold [63], with acyclic molecules as singleton groups; groups are shuffled with the split seed, sorted by decreasing size, and assigned to the partition whose size is lowest relative to its target fraction. Listing 2 gives the procedure as executed; the saved manifests, not the code, are what every model consumes. Unlike the DeepChem splitter used for MoleculeNet, large scaffold groups can fall into validation or test, so scores are not directly comparable with published scaffold split values. MoleculeNet recommends random splits for ESOL, FreeSolv, and Lipophilicity [8]; we used scaffold splits for ESOL and Lipophilicity to test generalisation beyond the training scaffolds, which is known to be harder but still optimistic relative to prospective use [19, 21].

The regression targets remain on their native scales. Source-level target summaries, class prevalence, and dataset provenance are provided in Additional file 1, Table S1. The fixed modelling settings are listed in Additional file 1, Table S2.

The five benchmark endpoints were modelled independently. Thus, this study evaluates a common multi-representation workflow across several tasks, rather than joint multi-task learning with parameters shared across multiple endpoints.

Reference models and external comparator

Reference models were a dummy predictor, an extended-connectivity fingerprint with radius 2 and 2,048 bits (ECFP4), RDKit 2D descriptors with a Random Forest (100 trees), and a three-layer descriptor MLP. The external comparator was Chemprop v2.3.1 [49] (hidden size 300, depth 3, up to 50 epochs with early stopping on validation loss), as a graph-only D-MPNN and with the 210 descriptors as molecule-level features (x_d). Five framework configurations isolated feature roles: `prior_only`, `direct_only` (descriptor MLP), `residual_only` (graph only), `direct_hybrid` (graph plus 210 descriptors), and `prior_residual` (frozen six-descriptor prior plus graph residual). Feature scalers, target scalers, and priors were fitted on training data only; hyperparameters were fixed a priori (Additional file 1, Table S2).

Evaluation and statistical analysis

The primary metrics were test root mean squared error (RMSE) for regression and area under the receiver operating characteristic curve (AUROC) for binary classification. Secondary metrics were mean absolute error (MAE), coefficient of determination (R^2), and average precision (AP), and per-split primary scores are given alongside them in Additional file 1 (Tables S13 and S14). Results are means and population standard deviations (SDs) over ten splits. Because methods share splits, comparisons use paired differences $\Delta_s = m_A(s) - m_B(s)$, summarised by their mean, standard deviation, win count, and a 95% confidence interval (CI) based on a t distribution with nine degrees of freedom. Overlapping training sets make such intervals optimistic [23, 22], so they are read descriptively, without multiplicity correction, and an interval that includes zero is treated as inconclusive rather than as equivalence [26]. Training-seed variance was estimated by repeating four configurations with three seeds on one BBBP split.

Three distribution-free checks accompany the intervals. First, every paired comparison was also tested by an exact sign-flip permutation test over the ten splits (all 2^{10} sign assignments enumerated, two-sided), with Holm correction inside the same comparison families; the permutation test does not assume normality, but like the naive t test it treats splits as exchangeable and is therefore reported as a secondary, more permissive view rather than as a replacement for the corrected interval. Second, the six selectors were compared as a randomised block design with dataset–budget cells as blocks: a Friedman omnibus test, mean ranks, and a Nemenyi critical difference at $\alpha = 0.05$ [24]. Third, the recommender was compared with random subsets using 50 independent draws per split and budget in every dataset, and the recommender's rank within that distribution was converted to a per-split p-value (resolution $1/51$) and combined across splits with Fisher's method [25]; this replaces the earlier comparison with a ten-draw mean, which carried the draw noise of the control.

Because the framework and the Chemprop comparator ran in different environments, the descriptor block was exported from both and compared cell by cell (`benchmarks/diagnose_descriptor_contract.py`). The two matrices agreed in 99.87% of 1,999,620 cells; the differences were confined to ten descriptors, mostly functional-group counts (`fr_unbrch_alkane`, `fr_prisulfonamd`, `NumHAcceptors`) and a small number of BCUT2D, `SPS` and `AvgIpc` values whose definitions or implementations changed between RDKit releases. The two implementations therefore saw effectively the same descriptor block, but not a bit-identical one; this is one reason the cross-implementation comparison is read as an independent replication of the descriptor effect rather than as an identical-input ablation.

No run was excluded from any table. The two runs that departed from their configuration, both on ESOL split 3, were diagnosed instead (`benchmarks/diagnose_esol_divergence.py`). The protocol had three parts: (i) describe the descriptor shift of every split in standardised units, (ii) retrain the affected route with additional training seeds, holding the split, descriptors, budget and all hyperparameters fixed (three seeds for the framework route, six for the Chemprop configuration), and (iii) rerun the Chemprop configuration with the standardised descriptor block winsorised at ± 5 training standard deviations, for the same six seeds. The third part is an intervention: if the divergence disappears when only the magnitude of the descriptor inputs is bounded, the cause is the input rather than the optimisation. All values in the main tables come from the original runs; the replications are used only to characterise the failure.

Sensitivity analyses

Two settings are conventions rather than quantities estimated from the data: the redundancy threshold of the greedy pruning step (0.85) and the encoder width of the routing experiments (128). Each was varied one factor at a time, with everything else fixed, and neither variation was used to choose the configuration reported in the main tables. The redundancy threshold was varied over 0.70–0.90 with $K = 20$ fixed, on ESOL, FreeSolv and BBBP, and the overlap between the selected sets was measured with the Jaccard index against the default threshold (Additional file 1, Table S20). Encoder width was varied over 64, 128 and 256 on FreeSolv splits 1–3, retraining the graph-only and direct-hybrid arms from scratch (Additional file 1, Table S21). These analyses test whether a conclusion survives a defensible change of a setting; they do not establish that any setting is optimal.

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